Substituting (1) in (3), we get $\frac{d^2y}{dx^2}+y=0$ as the required differential equation.

Example 10.4

Find the differential equation of the family of circles passing through the points $(a,0)$ and $(-a,0)$.

Solution

A circle passing through the points $(a,0)$ and $(-a,0)$ has its centre on y - axis.

Let $(0,b)$ be the centre of the circle. So, the radius of the circle is $\sqrt{a^2+b^2}$.

Therefore the equation of the family of circles passing through the points $(a,0)$ and $(-a,0)$ is $x^2+(y-b)^2=a^2+b^2$, b is an arbitrary constant. ... (1)

Differentiating both sides of (1) with respect to x, we get

$$2x+2(y-b)\frac{dy}{dx} = 0 \Rightarrow y-b=-\frac{x}{\frac{dy}{dx}} \Rightarrow b=\frac{x}{\frac{dy}{dx}}+y.$$

Substituting the value of b in equation (1), we get

$$x^2+\frac{x^2}{\left(\frac{dy}{dx}\right)^2} = a^2+\left(\frac{x}{\frac{dy}{dx}}+y\right)^2$$

$$\Rightarrow x^2\left(\frac{dy}{dx}\right)^2+x^2 = a^2\left(\frac{dy}{dx}\right)^2+\left[x+y\left(\frac{dy}{dx}\right)^2\right]^2$$

$\Rightarrow (x^2-y^2-a^2)\frac{dy}{dx}-2xy=0$, which is the required differential equation.

Example 10.5

Find the differential equation of the family of parabolas $y^2=4ax$, where a is an arbitrary constant.

Solution

The equation of the family of parabolas is given by $y^2=4ax$, a is an arbitrary constant. ... (1)

Differentiating both sides of (1) with respect to x, we get $2y\frac{dy}{dx}=4a \Rightarrow a=\frac{y}{2}\frac{dy}{dx}$

Substituting the value of a in (1) and simplifying, we get $\frac{dy}{dx}=\frac{y}{2x}$ as the required differential equation.

Example 10.6

Find the differential equation of the family of all ellipses having foci on the x-axis and centre at the origin.

Solution

The equation of the family of all ellipses having foci on the x-axis and centre at the origin is given by $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1,\ a>b$ ... (1)

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